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BELLOWS-L4: Superscalar LZ4 and Canonical Huffman Codec Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 11:58 UTC+8 Beijing Time)

Abstract

BELLOWS-L4 is a four-channel scatter-gather codec subsystem for lossless weight and KV-cache data transport. It preserves legacy LZ4 interfaces and adds an integrated 64-bit, four/eight-byte copy pipeline, hardware canonical Huffman coding, reversible BF16 exponent separation, W4 nibble coding and LZ4-to-Huffman post-coding. Software-vs-Hardware numerical comparison used independent Python models against simulated RTL: the LZ4 suite compared 2,294,234 bytes and the entropy suite 5,247,980 bytes without mismatches. On 32,768-byte checksum-free fixtures, integrated decompression accelerated literal-heavy/match-heavy workloads by 7.96x/5.98x with COPY_BYTES=4 and 15.79x/11.86x with COPY_BYTES=8 relative to supplied L3 RTL [E059, E060]. L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA achieved WNS_setup=+0.324 ns and hold slack=+0.038 ns at 10.0 ns on Arria 10 GX. L3 physical FPGA HIL passed three consecutive deterministic bridge-connectivity probes; these were not on-board codec workload or numerical-parity tests [E065–E067].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

bellows_l4_top instantiates four private contexts, a shared scheduler and SG DMA. Read, write and completion arbitration are independent. Contexts validate descriptors, route codecs, maintain queues and settle transactions before completion. Eight synchronous history banks and bounded overlap forwarding sustain full-width copy beats without a serial eight-byte dependency chain. Entropy construction scans registered memories, favoring bounded logic depth over low codebook latency. Mode 4 buffers the complete intermediate LZ4 block between coding phases. Separately elaborated byte RAMs improve synthesis scalability; replicated compressor literal storage preserves two independent synchronous reads and parent collision semantics [E001, E018–E026].

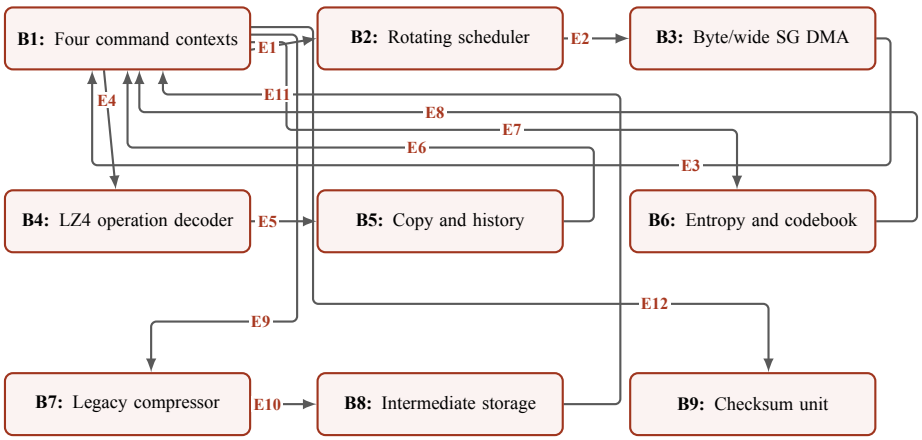


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Four command contexts	Per-channel command capture, descriptor validation, queues, codec routing, sticky faults and completion accounting.
B2: Rotating scheduler	Independent read, write and completion arbitration; captured ownership and payloads survive stalls.
B3: Byte/wide SG DMA	External memory boundary with one globally outstanding read and one write; responses return to captured owners.
B4: LZ4 operation decoder	Parses tokens, extensions and offsets; submits validated literal and match operations.
B5: Copy and history	Two registered stages, eight synchronous banks, pending-write forwarding and intra-beat overlap selection.
B6: Entropy and codebook	Hardware histogram/tree construction, canonical validation, bit packing and BF16/W4 transforms.
B7: Legacy compressor	Explicit adapter preserves the deterministic parent L2 compression algorithm and byte stream.
B8: Intermediate storage	Private banked buffer holds mode-4 LZ4 bytes; contexts control producer/consumer phase separation.
B9: Checksum unit	Per-context XXH32/CRC computation over original user bytes; results return to context framing and comparison logic.

Directed relationship	Meaning
E1: B1 to B2	Memory requests and completions
E2: B2 to B3	Selected requests and ownership
E3: B3 to B1	Read data and write acknowledgements
E4: B1 to B4	Compressed LZ4 input
E5: B4 to B5	Validated copy operations
E6: B5 to B1	Reconstructed beats
E7: B1 to B6	Entropy input and control
E8: B6 to B1	Encoded or decoded bytes
E9: B1 to B7	Raw compression input
E10: B7 to B8	Mode-4 LZ4 bytes via context
E11: B8 to B1	Buffered intermediate reads
E12: B1 to B9	Original-byte checksum stream

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock; active-low asynchronous architectural reset. RAM data processes are synchronous and do not reset arrays. Public outputs are zero during reset.
cmd_valid, cmd_ready and command fields	Input; cmd_ ready output	channel 2; id 16; decompress 1; checksum_ mode 2; src_ bytes/dst_ capacity 32 each; src_ segments/dst_ segments 5 each; wide 1; codec 3	Fields use the cmd_ prefix and latch only on valid/ ready. cmd_wide selects byte or wide transport per command. Four channels can remain live with different modes.
sg_valid, sg_ready, sg_ channel, sg_destination, sg_addr, sg_bytes	Input; sg_ ready output	1, 1, 2, 1, 64, 32 bits respectively	Collect source/destination segments for the selected admitted command. Positive lengths, exact totals and address-overflow checks precede memory traffic.
rd_req_*, rd_rsp_*	Request outputs/ready input; response inputs/ready output	channel 2; address 64; response data 8; controls/ error 1	Preserved byte-read transport. Each accepted request reads one byte; response ownership is implicit in the single outstanding transaction.
wr_req_*, wr_rsp_*	Request outputs/ready input; response inputs/ready output	channel 2; address 64; request data 8; controls/error 1	Preserved byte-write transport with explicit success/ error acknowledgement.
w_rd_req_*, w_rd_rsp_*	Request outputs/ready input; response inputs/ready output	channel 2; address 64; requested bytes 4; response data 64; controls/ error 1	Reads 1..COPY_BYTES contiguous bytes within one segment. Increasing addresses occupy increasing low byte lanes; unused response lanes are ignored.
w_wr_req_*, w_wr_rsp_ *	Request outputs/ready input; response inputs/ready output	channel 2; address/data 64; strobe 8; controls/error 1	Writes contiguous low lanes without crossing a destination segment. Unused data lanes are zero. Request acceptance alone does not count as successful output.

Signal / group	Direction	Width	Contract
status_valid, status_ ready and completion fields	Output; status_ready input	channel 2; id 16; error 4; src_bytes/dst_ bytes/ checksum 32 each; valid/ ready 1	Stable completion payload under backpressure. status_dst_bytes counts successful write acknowle dgements; status_src_bytes counts returned request bytes. err_checksum_mismatch is one bit and asserts only for a valid code-7 completion.
bellows_l4_copy_engine leaf	Operation inputs; result outputs	op_data/m_ data 64; op_ count/m_ count 4; op_ offset 16; handshake/ control 1	clk, rst_n, clear, op_valid/op_ready, op_match, op_ data, op_count, op_offset and m_valid/m_ready/m_ data/m_count form the standalone boundary. Operations are prevalidated and contain 1..COPY_ BYTES bytes.

Parameter	Default	Meaning / supported values
MAX_BLOCK_BYTES	65536	Integer 1..65536; bounds raw input and reconstructed output. Intermediate capacity is 2*MAX_BLOCK_ BYTES+16; entropy frame limit is 4*MAX_BLOCK_ BYTES+1024 excluding checksum.
SG_DEPTH	16	Integer 1..16 descriptors per source/destination list.
COPY_BYTES	8	Exactly 4 or 8; external wide data ports remain 64 bits. Invalid parameters execute a diagnostic containing PARAMETER ERROR.

III Behavior and corner cases

A Codec selection and legacy compatibility

cmd_codec selects raw LZ4 (0), byte Huffman (1), BF16 exponent Huffman (2), W4 nibble Huffman (3), or LZ4 followed by Huffman (4). cmd_decompress selects direction. Values 5..7 are invalid. Legacy L1/L2/L3 tops remain available. Entropy framing is BELLOWS-specific, not Zstandard-compatible.

B Copy timing and overlap

An operation accepted at edge E occupies the request/history-read stage after E; its output becomes valid after E+1 and can retire at E+2. Consecutive operations produce consecutive beats. Backpressure freezes upstream pipeline state without duplicate history writes. clear accepts no operation and starts fresh logical history only after prior work drains. Match bytes reference their own current output position minus offset, including earlier bytes in the same beat; offsets 1..65535 require available private history.

C Canonical entropy format

A 20-byte header contains BL4H, version 1, codec, two zero reserved bytes, then little-endian 32-bit N, S and B. Ascending-symbol length tables contain 256 bytes for modes 1/2/4 or 16 for mode 3. Mode 2 adds N/2 residual bytes; payload occupies exactly ceil(B/8) bytes. Codes sort by length then symbol and emit MSB-first; unused low padding bits are zero. Require S=N, N/2, or 2*N for modes 1, 2, or 3 respectively; mode 4 declares intermediate LZ4 length and must reconstruct exactly N.

D Hardware codebook rules

Repeated merges choose the smallest frequency/minimum-contained-symbol pair. Empty input uses zero lengths and B=0; a singleton uses one-bit code 0. Depth above 15 triggers a complete eight-bit byte alphabet or four-bit nibble alphabet table. Decoders also accept independently constructed complete canonical tables, rejecting oversubscription, forbidden incompleteness, invalid codes, inconsistent counts, truncation, extra bytes and bad padding.

E Exact BF16 and W4 transforms

For little-endian BF16 lo,hi: exponent=((hi&0x7f)<<1)|(lo>>7), residual=(hi&0x80)|(lo&0x7f). Reconstruction uses lo=(residual&0x7f)|((exponent&1)<<7), hi=(residual&0x80)|(exponent>>1). All exceptional encodings and NaN payloads remain unchanged. Odd-length BF16 encoding is rejected before memory traffic. W4 emits low then high nibble without arithmetic or requantization.

F Checksums

Mode 0 reports zero without a trailer. Mode 1 uses seed-zero XXH32 with modulo-2^32 arithmetic; mode 2 uses reflected CRC-32/ISO-HDLC, polynomial 0xedb88320 and initial/final XOR 0xffffffff. Both append four little-endian bytes covering original user data only. Checksum mode 3 is invalid.

G Transactions, reset and failure

Descriptors become eligible after command admission; memory waits for complete validation. Responses may handshake no earlier than the edge following request acceptance. Held requests and completions remain stable. Arbitration serves continuously eligible channels within four successful grants. Reset aborts ownership; the bridge must discard old responses. Completion waits for reserved transactions and output/checksum work to settle. First fault is sticky; simultaneous priority is bus, codec, block, capacity, checksum. Codes 0..11 mean success, truncated LZ4, zero offset, unavailable history, length overflow, block overflow, invalid stream usage, checksum mismatch, invalid command/descriptors, capacity overflow, memory error and malformed entropy. Prior acknowledged writes are not rolled back.

H Material implementation corrections

RAM-port normalization and separate byte-RAM leaves addressed mapping scalability while retaining synchronous old-data semantics. Descriptor remainder/beat-limit registers and parallel one-hot arbitration snapshots removed long request-control cones implicated in historical negative setup slack. A separate SG fixture error was corrected from mismatched descriptor totals to an exact 16-byte transaction; reference/status assertions were retained. Current measurements below supersede historical timing failures without erasing them.

IV Verification methods and measured results

A Software-vs-Hardware numerical agreement and compatibility

Method. Independent Python codec/checksum models compared against Icarus and compiled Verilator RTL; parent differential checks supplemented numerical expectations. Principal seeds were 17, 89 and 203 at both widths.

Result. LZ4: 1,650 transactions, 1,438 successful and 212 negative, with 2,294,234 compared bytes. Entropy: 1,496 transactions, 1,208 successful and 288 negative, with 5,247,980 compared bytes. No mismatches. Suite totals include directed/smoke traffic and are not unique-workload counts. Coverage includes 64-KiB cases, all BF16 encodings, both W4 positions and 144 malformed entropy cases per width.

Evidence: E059, E060, E032, E034

B Superscalar copy and integrated speedup

Method. Measured 4,096-byte steady-state copy streams; integrated tests measured accepted-command through accepted-status cycles for identical 32,768-byte blocks, earliest legal responses and no injected stalls.

Result. Copy sustained 4 and 8 bytes/cycle for literals and offsets 1,3,7,16, exceeding 0.99*COPY_BYTES. L3 literal/match cycles were 131,608/98,866; L4 width 4 used 16,527/16,529, giving 7.963x/5.981x against 4x thresholds. Width 8 used 8,336/8,337, giving 15.788x/11.859x against 8x thresholds. L4 throughput was approximately 1.983 and 3.931 bytes/cycle.

Evidence: E059

C Checksum and concurrent throughput characterization

Method. Repeated integrated fixtures with XXH32/CRC; measured four-channel mixed-codec, stalled traffic and reset recovery.

Result. XXH32 literal/match L4 cycles: 80,029/80,004 at width 4 and 77,981/75,902 at width 8. CRC cycles: 295,059/294,968 and 295,059/294,931. No-reset concurrent runs compared 4,482 bytes over 210,668/210,186 cycles, or 0.02128/0.02132 bytes/cycle. These mixed-workload rates are not peak LZ4 bandwidth. Arbitration maximum grant gap was four.

Evidence: E059

D Useful entropy compression and throughput

Method. Compared complete encoded frames against independent models, including headers/tables/residuals and excluding only outer checksum.

Result. For 8,192 raw bytes, repeated-byte/BF16-exponent-127/zero-W4 fixtures produced 1,300/4,884/2,084 bytes: 15.87%/59.62%/25.44%, passing 50%/70%/35% limits at both widths and all three seeds. Encode cycles were 69,723/56,923/117,771; decode cycles were 42,311/33,607/59,991. Table-unit tests performed 1,568 lookups across 12 fixtures, including depth-15 acceptance and hardware fallback.

Evidence: E060

E Defect sensitivity and structural synthesis

Method. Executed unchanged baseline assertions before six separately compiled mutants; synthesized the actual hierarchy with Yosys across all six boundary geometries.

Result. All six mutants were rejected by intended functional assertions: overlap, lane/tail, channel ownership, canonical bit order, BF16 residual and CRC polynomial. Synthesis retained memories and required hierarchy, with zero latches/division/modulo cells and successful driven-net consistency checks. Geometries were (1,1,4/8), (17,3,4/8) and (65536,16,4/8).

Evidence: E061, E062, E036

F L1 RTL timing lint

Method. Native hazard analysis of 23 delivered RTL files.

Result. Passed with zero combinational division/modulo, mixed-assignment, LUT clock-gating or AST loop-unroll hazards.

Evidence: E065

G L2 target timing and resources

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA of default bellows_l4_top at 100 MHz.

Result. Passed: WNS_setup = +0.324 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 103.35 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E066

H L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical PCIe bridge register-57 connectivity probes.

Result. All three passed deterministically and returned 178. The recorded tier is bridge-connectivity-only: no codec bitstream flashing, workload execution or FPGA numerical comparison is established.

Evidence: E067

I Clean portable reproduction

Method. Copied 50 declared sources into a fresh workspace and regenerated compact simulation, structural synthesis and source-portability checks.

Result. All three clean-workspace checks passed without original build products or prior evidence. This reproduced a compact subset, not the complete behavioral matrix or native hardware gates.

Evidence: E063

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 11	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

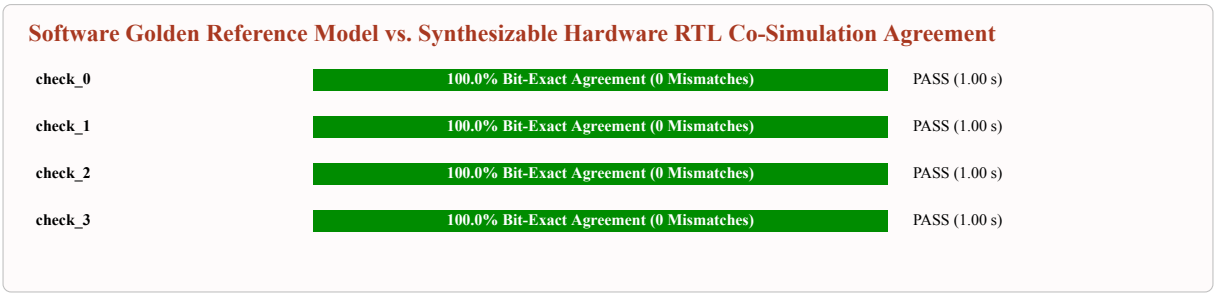


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.324 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

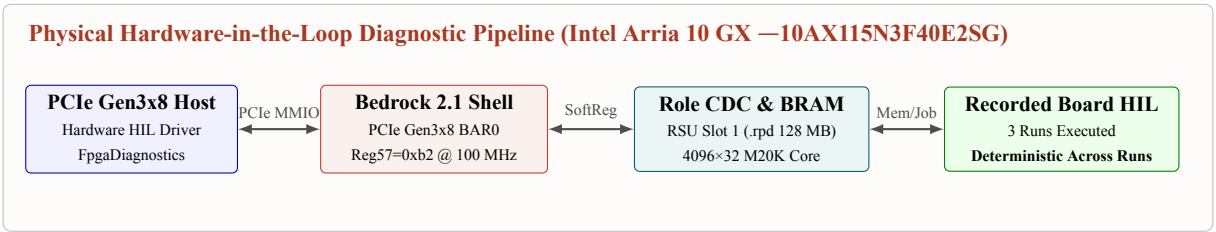


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/*.sv with top bellows_l4_top, including bellows_byte_ram.sv. Use design/SPEC.md for the frozen contract and design/ARCHITECTURE.md for routing and memory details. bellows_l3_reference/rtl/ remains comparison input, not a second implementation to compile into the same hierarchy.

Assert rst_n low, discard pre-reset responses, then release reset. Admit a command only on cmd_valid && cmd_ready; subsequently submit its declared descriptors. Source segments may be absent only for zero source length; destination capacity and segment count remain positive. Live-command source and writable destination regions must not alias.

Provide request backpressure and next-edge-or-later responses, independently for reads and writes. Honor segment-bounded counts/strobes and inspect accepted completion status before treating destination contents as valid.

From the project root, run python3 tests/verify.py for all suites, or select --suite lz4, entropy, defects, synthesis or portability. Dependencies are Python 3, Icarus 12.0/vvp, Verilator 5.020 and Yosys 0.33. Native timing and physical bridge execution remain external shared-runtime gates.

Detailed simulation ledgers are retained in evidence/l1-8981c398e5c94e56/receipt.json and evidence/l1-f9e418baad4e4815/receipt.json. Some source-document status paragraphs predate completed execution; use the current evidence-backed results above for measured scope.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 23 Verilog/SystemVerilog files; 3970 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/bellows_byte_ram.sv	22 total	20 code
rtl/bellows_l1_history.sv	30 total	27 code
rtl/bellows_l1_output.sv	65 total	61 code
rtl/bellows_l1_top.sv	307 total	281 code
rtl/bellows_l2_compressor.sv	281 total	268 code
rtl/bellows_l2_emitter.sv	185 total	183 code
rtl/bellows_l2_hash_match.sv	195 total	188 code
rtl/bellows_l2_literal_buffer.sv	53 total	51 code
rtl/bellows_l2_top.sv	61 total	58 code
rtl/bellows_l3_checksum.sv	130 total	130 code
rtl/bellows_l3_context.sv	314 total	310 code
rtl/bellows_l3_scheduler.sv	71 total	70 code
rtl/bellows_l3_sg_dma.sv	55 total	55 code
rtl/bellows_l3_top.sv	113 total	113 code
rtl/bellows_l4_context.sv	580 total	578 code
rtl/bellows_l4_copy_engine.sv	121 total	115 code
rtl/bellows_l4_entropy.sv	492 total	484 code
rtl/bellows_l4_huffman_table.sv	363 total	355 code
rtl/bellows_l4_intermediate.sv	64 total	64 code
rtl/bellows_l4_lz4_decoder.sv	195 total	190 code
rtl/bellows_l4_scheduler.sv	114 total	112 code
rtl/bellows_l4_sg_dma.sv	117 total	114 code
rtl/bellows_l4_top.sv	143 total	143 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E018	rtl/bellows_l4_context.sv
E032	tests/reference.py
E034	tests/scenarios.py
E036	tests/structural.py
E059	.vulcan/engineering/run-0001/attempt-0038/stdout.log
E060	.vulcan/engineering/run-0001/attempt-0039/stdout.log
E061	.vulcan/engineering/run-0001/attempt-0040/stdout.log
E062	.vulcan/engineering/run-0001/attempt-0041/stdout.log
E063	.vulcan/engineering/run-0001/attempt-0042/stdout.log
E065	.vulcan/engineering/run-0001/rtl-timing-lint.json
E066	.vulcan/engineering/run-0001/quartus-sta.json
E067	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).